SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	B/2024	**Date:**	21/10/26

Code of Conduct

I _____ *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula $s = \text{round}((L - 1) \times CDF)$, where $L = 256$. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3×3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3×3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Q1. Histogram Equalization

Given pixel values: 10, 10, 20, 20, 20, 30, 30, 40. Total pixels = 8 and L = 256.

Formula: CDF = cumulative frequency / total pixels; s = round((L − 1) × CDF) = round(255 × CDF).

Intensity	Frequency	CDF	New intensity
10	2	0.250	64
20	3	0.625	159
30	2	0.875	223
40	1	1.000	255

Equalized pixel values: 64, 64, 159, 159, 159, 223, 223, 255.

Comparison: The original intensities span 10–40, while the equalized values span 64–255. Histogram equalization spreads the available intensity range and improves global contrast.

Q2. Averaging and Median Filters

Given image: [[10, 20, 30], [20, 100, 20], [10, 20, 30]].

Averaging filter

Average = (10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30) / 9 = 260 / 9 = 28.89 ≈ 29.

Filtered image: [[10, 20, 30], [20, 29, 20], [10, 20, 30]].

Median filter

Sorted values: 10, 10, 20, 20, 20, 20, 30, 30, 100. The middle value is 20.

Median-filtered image: [[10, 20, 30], [20, 20, 20], [10, 20, 30]].

Comparison: The averaging filter is influenced by the outlier 100. The median filter replaces it with 20 and is generally better for removing impulse noise while preserving edges.

Q3. Laplacian Sharpening

Given centre pixel = 100 and its four direct neighbours are all 50.

Laplacian mask: [[0, −1, 0], [−1, 4, −1], [0, −1, 0]].

Laplacian response = 4(100) − (50 + 50 + 50 + 50) = 400 − 200 = 200.

Sharpened centre pixel = original value + response = 100 + 200 = 300.

For an 8-bit image, the value is clipped to the range 0–255, so the resulting centre intensity is 255.

Explanation: The Laplacian responds strongly to rapid intensity changes and emphasizes edges and fine details.

Q4. Global Thresholding

Given image: [[40, 60, 120], [80, 150, 200], [30, 100, 180]], threshold T = 100.

Rule: assign 1 when intensity ≥ 100; otherwise assign 0.

Binary image: [[0, 0, 1], [0, 1, 1], [0, 1, 1]].

Foreground pixels (1) = 5. Background pixels (0) = 4.

Explanation: Global thresholding separates brighter pixels from darker pixels using one threshold. It may fail when illumination varies across the image or foreground and background intensities overlap.

Q5. Sobel Edge Detection

Given image: [[10, 10, 10], [10, 50, 90], [10, 50, 90]].

Horizontal gradient (Gx)

Using the Sobel Gx mask: Gx = (−10 + 10) + (−20 + 180) + (−10 + 90) = 240.

Vertical gradient (Gy)

Using the Sobel Gy mask: Gy = (−10 − 20 − 10) + (10 + 100 + 90) = 160.

Gradient magnitude and direction

$G = \sqrt{(Gx^2 + Gy^2)} = \sqrt{(240^2 + 160^2)} = \sqrt{83200} \approx 288.44$.

Gradient direction $\theta = \tan^{-1}(Gy / Gx) = \tan^{-1}(160 / 240) \approx 33.69°$ from the positive x-axis.

Since Gx is larger than Gy, the horizontal intensity gradient is stronger, indicating a predominantly vertical edge. The edge orientation is perpendicular to the gradient direction.

Application: Edge information helps identify object boundaries and supports image segmentation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____